Stack Weights

Time limit: 1.00 s **Memory limit:** 512 MB

You have n coins, each of which has a distinct weight.

There are two stacks which are initially empty. On each step you move one coin to a stack. You never remove a coin from a stack.

After each move, your task is to determine which stack is heavier (if we can be sure that either stack is heavier).

Input

The first input line has an integer n: the number of coins. The coins are numbered $1, 2, \ldots, n$. You know that coin i is always heavier than coin $i-1$, but you don't know their exact weights.

After this, there are n lines that describe the moves. Each line has two integers c and s: move coin c to stack s (1 = left, 2 = right).

Output

After each move, print `<` if the right stack is heavier, `>` if the left stack is heavier, and `?` if we can't know which stack is heavier.

Constraints

- $1 \le n \le 2 \cdot 10^5$

Example

Input	Output
3 2 1 3 2 1 1	> < ?

Explanation: After the last move, if the coins are $[2, 3, 4]$, the left stack is heavier, but if the coins are $[1, 2, 5]$, the right stack is heavier.